

Streaming Data Analysis Administrative Items

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Me



Emanuele Della Valle, Ph.D.

prof @ Politecnico di Milano
founder @ Quantia Consulting
founder & CRO @ motus ml

25 years of experience in **research** and **innovation** projects
Expert in Semantic Technologies, Stream Computing, and Data Visualization

Brander of **Stream Reasoning**: an approach to tame the velocity and variety dimensions of Big Data simultaneously

Serial **startupper**:

<https://www.quantiaconsulting.com/>

<https://motusml.com/>



The other lecturers



Lorenzo Iovine, Dr.

PhD student @ Politecnico di Milano

Research in the field of **Streaming Machine Learning** for evolving multimodal data streams and concept drifts. Focus on applying Streaming Machine Learning techniques combined with embedding methods in industrial use cases.



Federico Giannini, Dr.

PhD student @ Politecnico di Milano

Researcher in the Streaming Machine Learning and **Continual Learning** fields. Focus on applying Deep Learning techniques to data streams to address concept drifts, temporal dependence, and catastrophic forgetting



Giacomo Ziffer, Dr.

PhD student @ Politecnico di Milano
founder & CEO @ motus ml

Researcher in the **Time-Evolving Analytics** field, focusing on applying Streaming Machine Learning techniques to (un)structured data streams with concept drifts and temporal dependence



The Course

- Public course **Web** page with the **official calendar of lecture**
 - <https://emanueledellavalle.org/teaching/streaming-data-analytics-2025-26/>
- Private **Webeep** page with **slides**, official **recordings**, and **announces**
 - <https://webeep.polimi.it/course/view.php?id=20659>
- Public **github** repo with all lectures' **code**
 - <https://github.com/Streaming-Data-Analytics/Courseware>



Lectures' Timetable

- **Official**

- Wednesday 14:15 - 16:15 in classroom 25.S.1
- Thursday 16:15 - 18:15 in classroom 2.1.2

- **Pragmatically**

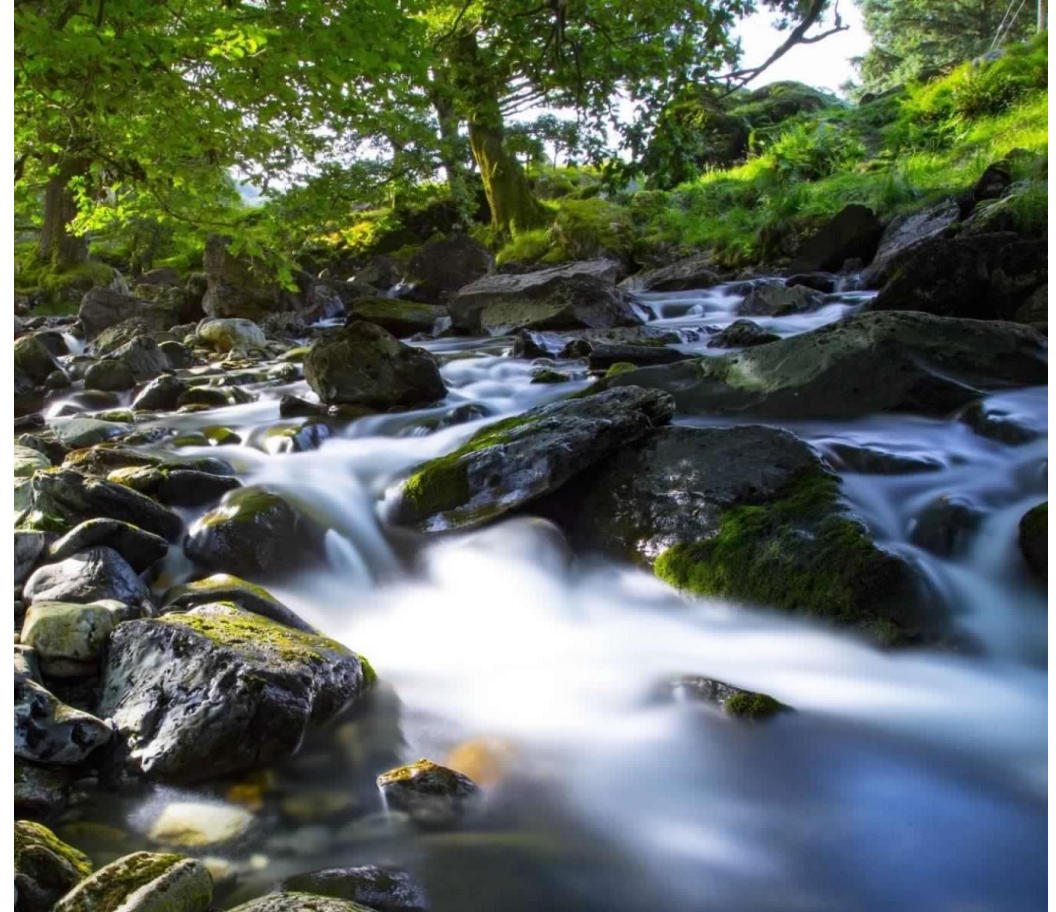
- Wednesday 14:30 - 16:00 in classroom 25.S.1
- Thursday 16:30 - 18:00 in classroom 2.1.2

- **Important NOTES**

- there is no clear cut between *theory* and *practice*
- *bring your laptops* we will often code
- I **record ALL lectures**, but I do not stream them

Objectives

- The course provides the foundational **concepts, methods, languages, and systems** for ingesting, processing, and analyzing **data that flows** to enable real-time decisions.
- The course covers the topics of
 - Streaming Data Engineering &
 - Streaming Data Science





Topics covered

Streaming Data Engineering

- The Streaming Data Engineering part of the course covers theoretical and practical aspects of Event-Based Systems (EBS), Data Stream Management Systems (DSMS), and Complex Event Processing (CEP).
- It uses the Event Processing Language ([EPL](#)) to illustrate typical DSMS and CEP operations. This segment equips students with essential skills for real-time data processing and analysis.
- It demonstrates horizontally scalable DSMS via [Apache Kafka](#), and [Apache Spark](#) and vertically scalable CEP via [Esper](#)



Topics covered

Streaming Data Science

- The Streaming Data Science part of the course covers the theoretical and practical aspects of Time Series Analytics (TSA), Streaming Machine Learning (SML), and Continuous Learning (CL), equipping students with essential skills for real-time data analysis.
- It demonstrates SML using River and MOA, TSA using statsmodels.tsa, and darts, and CL using Avalanche

Evaluation

- The exam consist of three parts
 - a **mandatory written exam** (max 30 marks)
 - an *optional continuous evaluations* (max 1 mark)
 - an *optional practical project work* with oral presentation (max 3 marks)

- **Example:**

- | | |
|-----------------------------------|------|
| • written exam | 27 + |
| • optional continuous evaluation | 1 + |
| • optional practical project work | 3 = |
| | 30L |



Evaluation

The "mandatory" written exam

- The written exam is composed of a mix of
 - **theoretical questions** regarding any course subjects and
 - **exercises** regarding the technical content and how to apply it in practice
- Students can get **up to 30/30** with the written exam



Evaluation

The "optional" continuous evaluation

- The optional continuous evaluation consists of quizzes proposed **during the lessons**
- They are based on **Wooclap**, offered by Politecnico di Milano as part of **Webeep**
- The quizzes are meant as a self-assessment
- Only in-person participation is allowed, remote participation is prohibited
- Students can get up to **1 mark** with continuous evaluation



Evaluation

The "optional" practical project

- The optional practical project requires students to use one or more of the technologies presented in the lectures to **solve a realistic** streaming data analytics **problem** based on data streams publicly available or provided by our industrial partners
- **Only students who will get at least 27/30 in the written exam can opt for it.**
 - The mark obtained with the optional continuous evaluation does not count.
- Students can get up to **3 marks** with the practical project

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